

Modular Operator Discovery TSP Report

Overview

- Condition count: 7.
- Best final transfer gap: phase7_modular_operator_random_replay.
- Best held-out TSPLIB gap: phase7_modular_operator_random_replay.
- Surviving modular candidates: none.

Run Metadata

- run_name: run_20260520_064300_p
- started_at_local: 2026-05-20 06:43:00
- finished_at_local: 2026-05-20 07:17:07
- duration_hhmm: 00:34
- duration_seconds: 2047.07
- seed_offset: 15000
- replicate_label: p
- judge_status: enabled
- judge_provider: openai
- judge_model: gpt-4.1-mini

Condition Comparison

Condition	Mode	Replay	Selection	Final TSPLIB Gap	Final Family Gap	Final Transfer Gap	Mean Novelty	Mean Complexity	Surviving Candidate	Transplant Delta	Pareto Runtime Inflation
phase7_baseline_heuristic_only	baseline_only	none	score_only	0.235058	0.054968	0.121317	0.0	0.0	False	0.0	0.0
phase7_full_solver_evolution	full_solver	none	score_only	0.213387	0.064916	0.159398	0.903718	0.56	False	0.0	0.0
phase7_modular_operator_evolution	modular_operator	none	score_only	0.235058	0.046321	0.115856	0.862441	0.445	False	0.009454	0.790432
phase7_modular_operator_random_replay	modular_operator	random	score_only	0.174128	0.074362	0.111118	0.0	0.445	False	-0.036453	1.105281
phase7_modular_operator_diversity_residual_replay	modular_operator	diversity_residual	score_only	0.235058	0.049737	0.118013	0.821465	0.445	False	0.009816	0.825788

Condition	Mode	Replay	Selection	Final TSPLIB Gap	Final Family Gap	Final Transfer Gap	Mean Novelty	Mean Complexity	Surviving Candidate	Transplant Delta	Pareto Runtime Inflation
phase7_modular_operator_compression_pressure	modular_operator	none	novelty_gate	0.235058	0.050827	0.118702	0.862626	0.44	False	0.010373	0.821431
phase7_modular_operator_pareto_selection	modular_operator	none	pareto	0.255674	0.119719	0.169808	0.0	0.4425	False	0.042452	1.679113

Condition Notes

phase7_baseline_heuristic_only

- Execution mode baseline_only on host scaffold nearest_neighbor_2opt.
- Final gaps: TSPLIB 0.235058, family holdout 0.054968, combined 0.121317.
- Accepted novelty 0.0, complexity 0.0, and adaptation efficiency 0.0.
- Last-epoch runtime 0.0 ms and distance evaluations 0.0.
- Validation: surviving False, transplant delta 0.0, positive scaffolds 0, Pareto runtime inflation 0.0.

phase7_full_solver_evolution

- Execution mode full_solver on host scaffold whole_solver.
- Final gaps: TSPLIB 0.213387, family holdout 0.064916, combined 0.159398.
- Accepted novelty 0.903718, complexity 0.56, and adaptation efficiency -0.027558.
- Last-epoch runtime 0.0 ms and distance evaluations 0.0.
- Validation: surviving False, transplant delta 0.0, positive scaffolds 0, Pareto runtime inflation 0.0.

phase7_modular_operator_evolution

- Execution mode modular_operator on host scaffold nearest_neighbor_2opt.
- Final gaps: TSPLIB 0.235058, family holdout 0.046321, combined 0.115856.
- Accepted novelty 0.862441, complexity 0.445, and adaptation efficiency 0.034424.
- Last-epoch runtime 116.588075 ms and distance evaluations 5381.5.
- Validation: surviving False, transplant delta 0.009454, positive scaffolds 1, Pareto runtime inflation 0.790432.
- Validation summary: {"better_gap_same_runtime": false, "code_length_lines": 12, "complexity_score": 0.44, "distance_eval_inflation": 0.856118, "gap_delta": -0.002615, "runtime_inflation": 0.790432, "same_gap_faster": false}

phase7_modular_operator_random_replay

- Execution mode modular_operator on host scaffold nearest_neighbor_2opt.
- Final gaps: TSPLIB 0.174128, family holdout 0.074362, combined 0.111118.
- Accepted novelty 0.0, complexity 0.445, and adaptation efficiency 0.0.
- Last-epoch runtime 68.09175 ms and distance evaluations 3810.75.

- Validation: surviving False, transplant delta -0.036453, positive scaffolds 4, Pareto runtime inflation 1.105281.
- Validation summary: {"better_gap_same_runtime": false, "code_length_lines": 13, "complexity_score": 0.46, "distance_eval_inflation": 11.887779, "gap_delta": -0.014035, "runtime_inflation": 1.105281, "same_gap_faster": false}

phase7_modular_operator_diversity_residual_replay

- Execution mode modular_operator on host scaffold nearest_neighbor_2opt.
- Final gaps: TSPLIB 0.235058, family holdout 0.049737, combined 0.118013.
- Accepted novelty 0.821465, complexity 0.445, and adaptation efficiency 0.009255.
- Last-epoch runtime 139.094125 ms and distance evaluations 4770.25.
- Validation: surviving False, transplant delta 0.009816, positive scaffolds 1, Pareto runtime inflation 0.825788.
- Validation summary: {"better_gap_same_runtime": false, "code_length_lines": 12, "complexity_score": 0.44, "distance_eval_inflation": 0.892165, "gap_delta": -0.005462, "runtime_inflation": 0.825788, "same_gap_faster": false}

phase7_modular_operator_compression_pressure

- Execution mode modular_operator on host scaffold nearest_neighbor_2opt.
- Final gaps: TSPLIB 0.235058, family holdout 0.050827, combined 0.118702.
- Accepted novelty 0.862626, complexity 0.44, and adaptation efficiency 0.016828.
- Last-epoch runtime 75.22365 ms and distance evaluations 4892.0.
- Validation: surviving False, transplant delta 0.010373, positive scaffolds 1, Pareto runtime inflation 0.821431.
- Validation summary: {"better_gap_same_runtime": false, "code_length_lines": 12, "complexity_score": 0.44, "distance_eval_inflation": 0.869479, "gap_delta": -0.003304, "runtime_inflation": 0.821431, "same_gap_faster": false}

phase7_modular_operator_pareto_selection

- Execution mode modular_operator on host scaffold nearest_neighbor_2opt.
- Final gaps: TSPLIB 0.255674, family holdout 0.119719, combined 0.169808.
- Accepted novelty 0.0, complexity 0.4425, and adaptation efficiency 0.0.
- Last-epoch runtime 71.39275 ms and distance evaluations 4515.0.
- Validation: surviving False, transplant delta 0.042452, positive scaffolds 0, Pareto runtime inflation 1.679113.
- Validation summary: {"better_gap_same_runtime": false, "code_length_lines": 13, "complexity_score": 0.46, "distance_eval_inflation": 2.049916, "gap_delta": 0.042667, "runtime_inflation": 1.679113, "same_gap_faster": false}

Judge Appendix

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TSP Modular-Operator Discovery Suite: Summary & Interpretation

Main Transfer/Generalization Evidence (Held-out TSPLIB & Family Holdout Gap)

- **Held-out TSPLIB gap:**
- Baseline heuristic (phase7_baseline_heuristic_only): 0.235058
- Operators with modular execution modes show no improvement or slightly worse gaps on held-out TSPLIB instances (all around 0.174-0.235, none better than baseline).
- Best modular operator condition on TSPLIB gap:
- phase7_modular_operator_random_replay achieves 0.174128 but suffers from ablation and transplant weaknesses (see below).
- **Family holdout gap (family gap):**
- The smaller final family gap relative to baseline (0.054968) indicates potential transfer; modular operators sometimes show modest improvements or slight regressions.
- Best family gap results are from restart controllers (phase7_modular_operator_evolution, phase7_modular_operator_diversity_residual_replay, phase7_modular_operator_compression_pressure) showing ~0.046–0.050, slightly better than baseline heuristic at 0.054968.
- Scaffold selectors show mixed or worse family gap results (sometimes severe regressions).

Operator Validation: Transplant & Ablation Checks

Restart Controllers (restart_controller type)

- **Ablation:**
- Ablation gap deltas near zero (≤ 0.005) indicating operator effect is small but present (restarts contribute small but consistent gap gains).
- Variants disabling the restart operators increase gap slightly, consistent with a modest real effect.
- **Transplant:**
- Transplant mean gap deltas mostly positive ($\sim +0.009$ to $+0.010$), indicating transplant hurts performance overall.
- Positive scaffold transplant count is low (1 per restart operator), transplanting only cleanly into the native scaffold (nearest_neighbor_2opt) or few others.
- Does not transplant cleanly into sparse_three_opt or clustered_local_search.
- **Runtime Inflation:**
- Significant runtime inflation (~ 0.82 to 0.83), which is a downside.
- **Overall:**

Restart controllers provide **some transferable operator structure** (modest gap reduction on family holdout + stable ablation), but transfer beyond native scaffold is weak, with runtime costs and transplant issues. No surviving candidate.

Scaffold Selectors (scaffold_selector type)

- Example: phase7_modular_operator_random_replay (best TSPLIB performer) and phase7_modular_operator_pareto_selection:
- **Ablation:**

- Disabling operator does not hurt or even improves performance in some variants (e.g., flattened_selector).
- Some variants (shuffled selector) improve gap further, suggesting core operator effect is not robust or clear.
- **Transplant:**
- Varying transplant gap delta:
 - random_replay selection shows mean transplant gap delta -0.036 (positive effect in transplant) with 4 positive scaffolds.
 - However, severe regressions on several families balance this out.
 - Other scaffold selectors show negative transplant results and do not transplant cleanly in multiple scaffolds.
- **Family & Held-out Gaps:**
- Mixed signals: underperforms in several important families (e.g., heldout_tsplib, grid_like_tsp, nearest_neighbor_trap_tsp).
- Positive signal mainly in two_cluster_bottleneck_tsp but not uniformly.
- **Runtime Inflation:**
- High runtime inflation (>1.0, up to 1.68 in pareto selection), a negative tradeoff.
- **Overall:**

Scaffold selectors fail ablation validation (disabling does not hurt), show mixed or poor transfer with multiple severe regressions, and suffer high runtime inflation. No robust reusable operator structure demonstrated.

Pareto Tradeoffs

- None of the modular operators simultaneously improve gap and runtime over baseline.
- Pareto gap deltas are marginal negative (small gap reductions) or even positive (worse gap), while runtime inflation consistently high (≥ 0.79 for restart controllers, > 1.0 for scaffold selectors).
- No condition shows "better gap at same runtime" or "same gap faster."

Surviving Candidates

- **None** survived all checks including transplant, ablation, and family holdout tests.
- The modular restart controllers show the most promise with modest positive family holdout gap improvement and some ablation evidence but fail transplant cleanly and suffer runtime cost.
- Scaffold selectors have inconsistent benefit and fail ablation validation.

Conservative Conclusions

1. **No modular operator qualifies as a surviving, fully transferable reusable operator** given the failure in transplant and ablation checks, plus runtime inflation.

2. ****Restart controllers (adaptive restart schedules) show modest validation support****:
- They yield small consistent improvements on family holdout gap (~0.046 vs 0.055 baseline).
 - Ablation shows some operator effect.
 - However, poor transplant performance, runtime inflation (~82%), and failure on key scaffold transplants limit claims.

This supports ****reusable operator structure related to adaptive restart control in nearest_neighbor_2opt and closely related scaffolds only****, but no broad transfer or solver discovery.

3. ****Scaffold selectors exhibit unstable performance with ablation failing to show clear operator effect****, have multiple severe regressions on families, and runtime overhead. Transfer and transplant is inconsistent. No clear reusable abstraction.
4. ****Full-solver evolution and baseline remain the strongest references****:
- Neither baseline nor full-solvers produce reusable modular operators per criteria.
 - The modular operator conditions do not clearly outperform baseline on held-out TSPLIB gap.
5. ****Best performing condition on held-out TSPLIB gap, phase7_modular_operator_random_replay, does not survive ablation nor transplant fully; thus no claim of real reusable discovery.****

****Summary Table****

Operator Type	Best Family Gap	Held-out TSPLIB Gap	Ablation Valid?	Transplant Valid?	Runtime Inflation	Surviving Candidate?	Notes
Restart Controllers	~0.046 (vs 0.055 baseline)	~0.235 (same as baseline)	Modest +	No (fails cleanly on multiple scaffolds)	~0.82	No	Reusable restart structure in native scaffold only; runtime cost
Scaffold Selectors	Mixed (some severe regressions)	~0.174 best (random replay) but with ablation failures	No	Mixed; some transplant benefit but severe regressions	High (>1.0)	No	Operator effect unclear; poor robustness
Full-solver (non-modular)	N/A	~0.213	N/A	N/A	Low	No	Baseline for comparison
Baseline Heuristic Only	0.055	0.235	N/A	N/A	N/A	N/A	Base reference

****Final Interpretation****

- The suite has ****not discovered a validated modular operator that generalizes robustly across held-out TSPLIB and family holdouts with surviving ablation and transplant checks****.
- Restart controllers offer a ****partially reusable operator pattern related to adaptive restart control in specific scaffold contexts****, but lack broad transfer or pareto improvements.
- Scaffold selectors lack robust modular operator evidence; ablation and transfer analyses suggest no true operator effect.
- No full solver algorithm discoveries from these modular operators under the strict criteria.

****Recommendation****

- Focus further research on improving restart controller transplantability and runtime efficiency to enhance practical transfer.
- Reconsider scaffold selector designs to produce clearer ablation signals and across-family generalization.
- Maintain baseline and full-solver evolutions as important performance baselines, as no modular operator yet overcomes them consistently.